

Review Comments

[1] Reframe the manuscript as a journal research article rather than a book chapter, because the abstract, Section 1, and Section 7 repeatedly use the word **chapter** and the current structure does not yet meet the evidential depth expected in a Q1 journal submission.

[2] State a precise research problem, research questions, and testable objectives before presenting the framework, because Sections 3.1–3.4 currently describe broad cybersecurity needs but do not define the exact problem that the proposed system solves.

[3] Clarify the novelty of the HoneyBee Method by comparing it directly with existing swarm, multi-agent, moving-target defense, and honeypot-based cyber defense methods, because the manuscript currently claims inspiration from honeybees without showing which technical mechanism is new.

[4] Define the threat model in a dedicated subsection, including protected assets, network topology, attacker knowledge, attacker capabilities, attack stages, assumed defender visibility, and the boundary between detection, deception, and response.

[5] Add a formal system model that specifies the agents, environment, observations, messages, actions, decision points, and feedback loops, because the present architecture is described only at a high level and cannot be reproduced from the current text.

[6] Map each honeybee-inspired concept to a concrete cybersecurity function, for example scout, worker, guard, waggle-dance communication, and colony learning, because the biological analogy is currently descriptive and not operationalized in the algorithmic design.

[7] Resolve the inconsistency between Figure 2 and Section 4.2, because the text claims four layers named Detection, Deception, Response, and Learning, while the figure shows Data Collection, Analysis and Detection, Decision and Response, and Action layers without a visible Learning layer.

[8] Revise Figure 2 so that all modules discussed in the manuscript are visible and consistently named, including adaptive honeypots, reinforcement learning, multi-agent coordination, learning feedback, and threat intelligence storage.

[9] Define the role of each agent type with inputs, outputs, local model, shared memory, communication protocol, and failure handling, because Section 4.7 mentions detection, deception, response, and learning agents but does not explain how they coordinate or avoid conflicting actions.

[10] Specify the machine-learning and deep-learning components used in the detection layer, including feature sets, preprocessing, model architectures, hyperparameters, class handling strategy, training procedure, and update schedule.

[11] Define the reinforcement learning or multi-agent reinforcement learning formulation with state space, action space, reward function, policy update method, exploration strategy, safety constraints, and termination conditions, because Section 4.5 currently states that reinforcement learning is used but does not provide the technical design.

- [12] Explain how the adaptive honeypot mechanism works in implementable detail, including redirection rules, decoy service selection, interaction logging, attacker engagement criteria, honeypot reconfiguration policy, and resistance to honeypot fingerprinting.
- [13] Replace the unexplained term ‘Corneya’ in Table 2 with the exact honeypot platform or module name, and describe its configuration, because this term is central to the reported deception engagement result but is not defined anywhere in the manuscript.
- [14] Provide complete dataset details in Section 5.1, including dataset names, versions, record counts, feature counts, attack classes, class distribution, train-validation-test split, preprocessing steps, and whether any synthetic traffic was generated.
- [15] Describe the experimental testbed in sufficient detail, including hardware, software, SDN controller, virtual machines, network services, traffic generation method, attack tools, scenario duration, and number of independent runs.
- [16] Rebuild Table 2 using measured experimental results rather than target-based status labels, because columns such as ‘Target/Benchmark’ and ‘Achieved’ make the evaluation appear self-confirming instead of independently benchmarked.
- [17] Define all evaluation metrics mathematically or operationally, including detection accuracy, false positive rate, initial containment time, full mitigation time, mean time to respond, and deception engagement, because the current manuscript reports values without explaining how they were computed.
- [18] Report detection results with precision, recall, F1-score, ROC-AUC or PR-AUC, confusion matrix, and per-attack-class performance, because accuracy and false positive rate alone are insufficient for intrusion detection datasets with possible class imbalance.
- [19] Add statistical reliability analysis, including repeated runs, confidence intervals, standard deviations, and significance testing against baselines, because the current single-value results do not show whether the observed improvements are stable.
- [20] Replace the comparative benchmarking in Table 3 with a fair experimental comparison under the same dataset, preprocessing, metrics, and hardware conditions, because the current comparison mixes unclear system labels and unsupported values from prior work.
- [21] Justify the baselines carefully by selecting current and relevant methods such as standalone IDS, deep IDS, RL-based response, static honeypot, moving-target defense, and multi-agent defense variants, and explain why each baseline is suitable for comparison.
- [22] Add ablation experiments that remove or isolate the main components, including no deception, no reinforcement learning, no learning feedback, single-agent control, and detection-only variants, because this is necessary to prove which part of the framework causes the reported gains.
- [23] Evaluate robustness under realistic conditions, including unseen attack types, concept drift, adversarial evasion, noisy traffic, delayed agent communication, agent failure, and honeypot discovery, because the proposed system claims adaptability but does not test it

directly.

[24] Clarify whether the reported framework has been implemented or is mainly conceptual, because the manuscript presents precise numerical KPIs but does not provide enough implementation evidence to support those values.

[25] Align the conclusion with the evidence actually provided, because claims such as closed-loop self-improvement, resistance to known and new attack patterns, and reduced manual intervention require stronger empirical support than the current results section provides.

[26] Strengthen the literature review by replacing broad entries such as ‘Honeypot studies (IEEE)’ and ‘RL-IDS studies’ with specific peer-reviewed papers, and synthesize the technical gaps rather than listing general findings.

[27] Verify every reference and complete missing bibliographic details, because several citations in Table 1 and the reference list appear too broad or difficult to trace, which can seriously affect credibility in a Q1 journal review.

[28] Remove the ‘Proposed framework’ row from the literature review table or clearly separate it from prior work, because placing the authors’ own 2025 framework inside the review table can confuse readers and weaken the objectivity of the related work.

[29] Standardize terminology across the manuscript, especially ‘HoneyBee Method’, ‘BIA’, ‘ABHEDYA’, ‘Detect-Reveal-Neutralize-Learn’, and ‘Detect-Mislead-Neutralize-Learn’, because inconsistent naming makes the framework difficult to follow.

[30] Improve the research gap section by explicitly linking each identified limitation in prior work to a corresponding design element in the proposed framework, because the current gap statement is repetitive and does not clearly justify the proposed integration.

[31] Add a complete algorithmic description or pseudocode for the full operational workflow from event monitoring to detection, deception, response, learning, and model update, because the current workflow remains narrative and cannot be independently implemented.

[32] Discuss deployment risks and governance issues, including privacy of collected attacker interaction logs, legal limits of deception, risk of blocking legitimate users, rollback of autonomous actions, and human approval for high-impact responses.

[33] Revise the results discussion so that each numerical claim is supported by a clear experimental observation, table, or scenario, because Sections 5.2–5.7 currently repeat broad claims without enough evidence-based interpretation.